Viktor Cervantes Gonzalez

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EDUCATION

University of California, Berkeley Graduation Date: May 2028

Bachelor of Arts: Data Science, Bachelor of Arts: Economics

Relevant Coursework/Languages: Java, Python, SQL, R, Data Structures & Algorithms, Structure and Interpretation of Computer Programs, Econometrics , Linear Algebra+Differential Equations, Discrete Mathematics. Advanced Probability, Financial Economics

PROFESSIONAL EXPERIENCE

EPC Power Corp. |Technical Project Manager August 2026 – Present

Computer Vision/ Machine Learning *Berkeley, CA*

- Leading a 6-person team building a computer vision defect detection system for PCB manufacturing (solder bridge/cable insert defects), serving as primary liaison to the client's engineering team
- Overseeing development of a containerized Docker inference service with asynchronous REST job processing, and plan the AWS infrastructure for production deployment
- Managing direct training/evaluation of YOLO object detection models on precision, recall, mAP, and inference performance

ConflictQuery | SWE Intern August 2026 – Present

LLM Engineering/ Full Stack Development *Berkeley, CA*

- Built and deployed a serverless **LLM app** (**Next.js**, **TypeScript**, **Vercel**) classifying governance scenarios via **Gemini API** schema-enforced output grounded in 129 policy sections, logging **per-request validation metrics** to **Firestore**
- Reduced **inference latency by over 60%** through model configuration tuning, and cut worst-case failure recovery from **63s to 7.5s** with jittered **retry backoff**
- **Benchmarked** context caching vs. structured output on **cost/latency**, documented in an **architecture decision record**

Aflac | Technical Project Manager January 2026 – May 2026

Data Engineering, OCR & Analytics *Berkeley, CA*

- Led enterprise data engineering engagement with **Aflac**, managing project scope, timelines, and delivery for a **6-person cross-functional team** of engineers and analysts
- Owning end-to-end delivery of a repeatable data ingestion and normalization **ETL pipeline** to extract and tabularize 5+ years of **U.S. Department of Labor Form 5500 filings,** producing a queryable dataset with 1M+ records
- Defined project requirements, success metrics, and data scope in collaboration with Aflac stakeholders
- Implemented **quality validation** and **error analysis workflows**

Egen AI | Data Engineer Consultant September 2025 – January 2026

ML Engineering, Applied Analytics & Platform Development *Berkeley, CA*

- Engineered and deployed an end-to-end hybrid machine learning solution on **Google Cloud Platform (GCP)** to **automate** project role allocation and **effort estimation**, processing both unstructured text and structured historical data
- Designed a **scalable data pipeline** using **BigQuery** and **Cloud Functions,** developed a feature engineering framework, and trained/evaluated multiple models (including **Transformers** and **XGBoost**) on **Vertex AI** to **predict** staffing needs.
- Led the full project lifecycle, **data ingestion** and **model selection** (optimizing **MAPE** and **interval scores**) to final migration, documentation, and technical knowledge transfer, delivering a **cost-effective** and **scalable production-ready system.**

PROJECTS

Japan Equity Research Pipeline | Python, Anthropic API, BeautifulSoup, pdfminer

- Built a 4-stage **machine learning pipeline** scraping Tokyo Stock Exchange same-day filings (**TDnet/JPX**) for 8 Japanese equity positions, processing Japanese-language PDFs natively across 90-day disclosure windows
- Engineered a **two-model architecture** using Claude Haiku for relevance triage and Claude Sonnet for deep thesis-impact analysis, **reducing API cost** ~3x vs. naive approach
- Delivered automated equity research reports with per-position confidence scores and **cross-position risk synthesis**, backed by **three-layer incremental caching** and structured **JSON** output for reproducibility and auditability

NFL QB Predictive Analytics Dashboard | Python, Scikit-learn, Streamlit, Plotly

- Developed end-to-end machine learning pipeline using **Scikit-learn** to predict college QBR, implementing **Linear, Ridge, KNN regression** with **GridSearchCV** hyperparameter optimization and **5-fold cross-validation**
- Engineered **predictive features** from raw college statistics including **EPA-per-play metrics** and **efficiency ratios**, applying **StandardScaler normalization** and **median imputation** for robust preprocessing
- Built interactive **Streamlit dashboard** with **Plotly visualizations** featuring real-time predictions, feature importance analysis, model **performance metrics** (MSE, R^2), with **95% confidence intervals**
- Automated NFL statistics collection via **ESPN API integration** and **web scraping** (BeautifulSoup, Requests)

Programming:: Python, Java, SQL, R, Pandas, NumPY, Typescript
ML & Analytics: Scikit-learn, XGBoost, Transformers, Feature Engineering, Model Evaluation, YOLO,
Cloud & Infrastructure: Google Cloud Platform (BigQuery, Vertex AI, Cloud Functions), AWS(S3), Docker, Vercel, Firestore
Data & Tools: Pandas, NumPy, Data Pipelines, ETL, Performance Benchmarking, Streamlit, Plotly, Git, GitHub, Excel
Languages : Spanish (native fluency), English
Outside Interests: Rock Climbing, Salsa Dancing, Biking, Table Tennis